

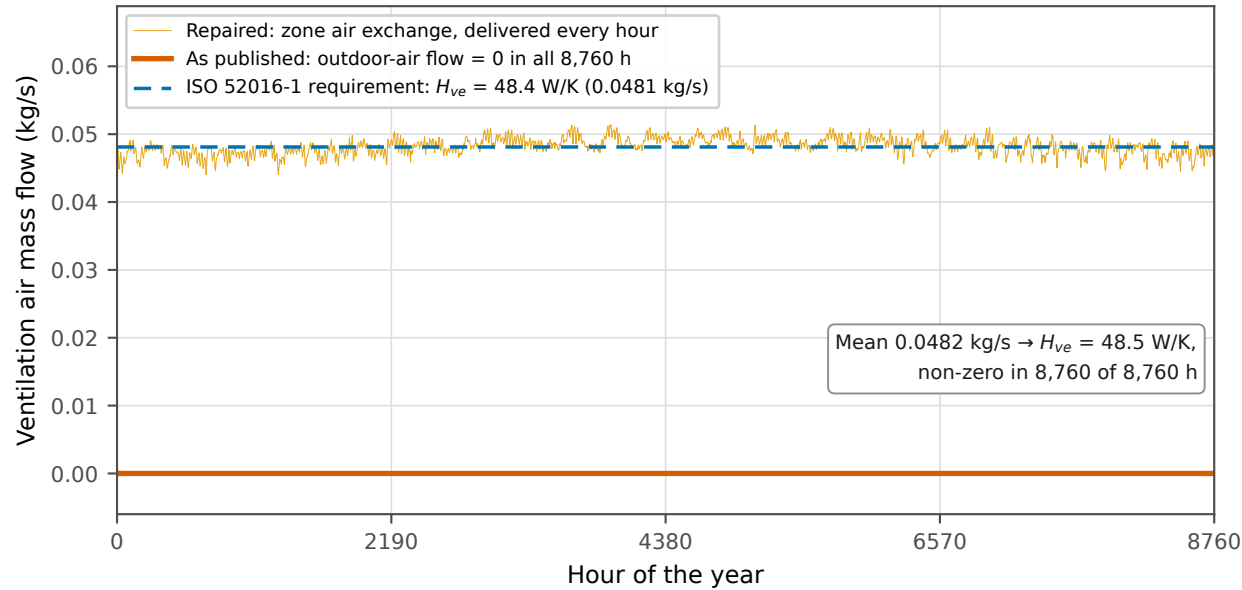
F11 — How the four defects in the EnergyPlus reference model were found

Each has a visible signature in an hourly series; none of the four is visible in an annual total

Hourly EnergyPlus 24.1.0 output, regenerated from the committed IDF with output variables added and nothing else changed; every run reproduces the committed annual totals before its series are drawn.
Only defect 4 has a committed defective input, so panel 4 is instrumented in the defective state and panels 1-3 in the repaired state, with the defective reading marked as an explicit gap.

1 Zero designed ventilation

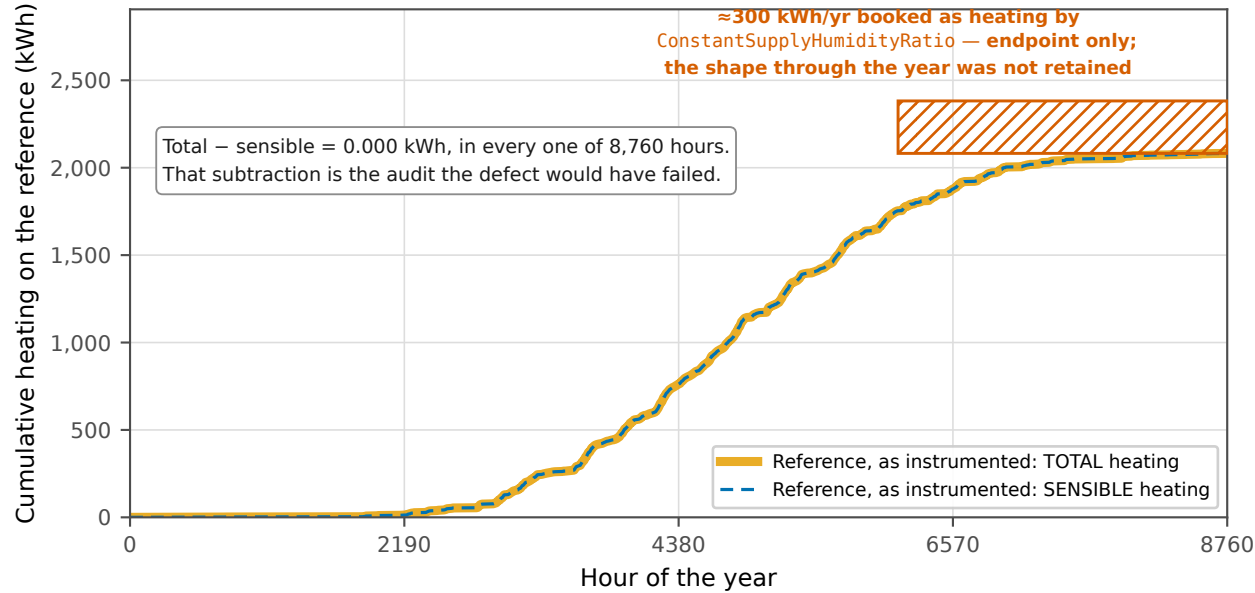
repaired state instrumented



Consequence: repairing defects 1-3 moves the reference's heating from 1,120.24 to 2,081.97 kWh, +961.73 kWh (+85.8 %). The three are not separated anywhere in the committed results, so no share of that is attributed to any one of them.

3 Humidification booked as heating

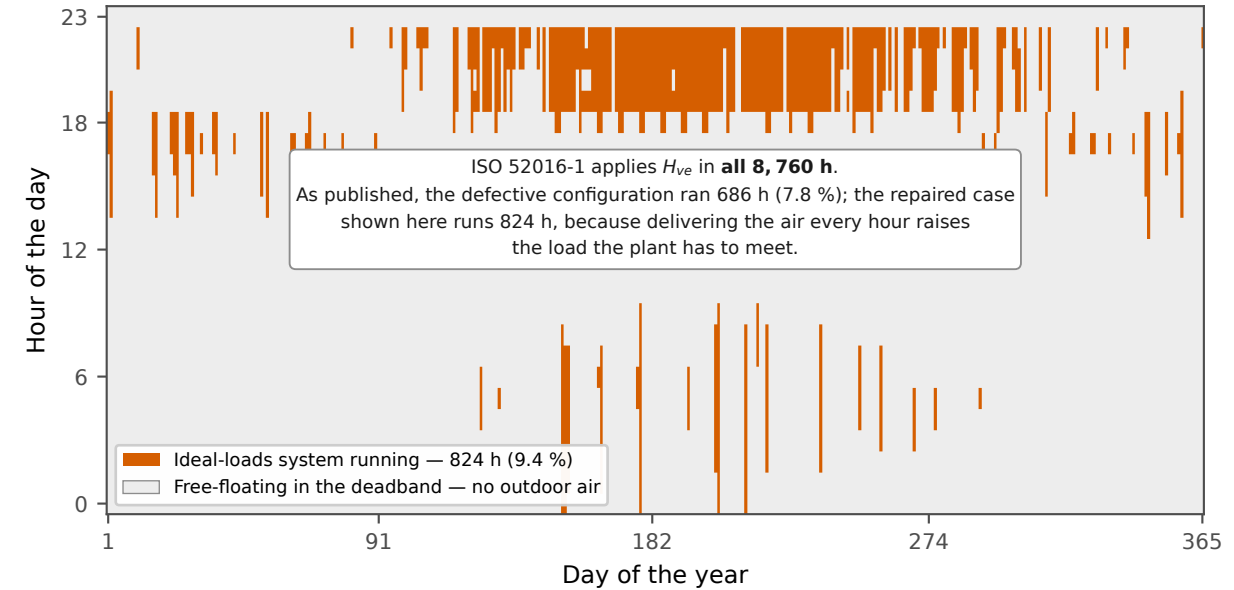
repaired state instrumented



Consequence: ≈300 kWh/yr of the same total. Masked by defect 1 — with no air flowing there was nothing to humidify — so it appeared only once defects 1 and 2 were fixed.

2 Intermittent outdoor air

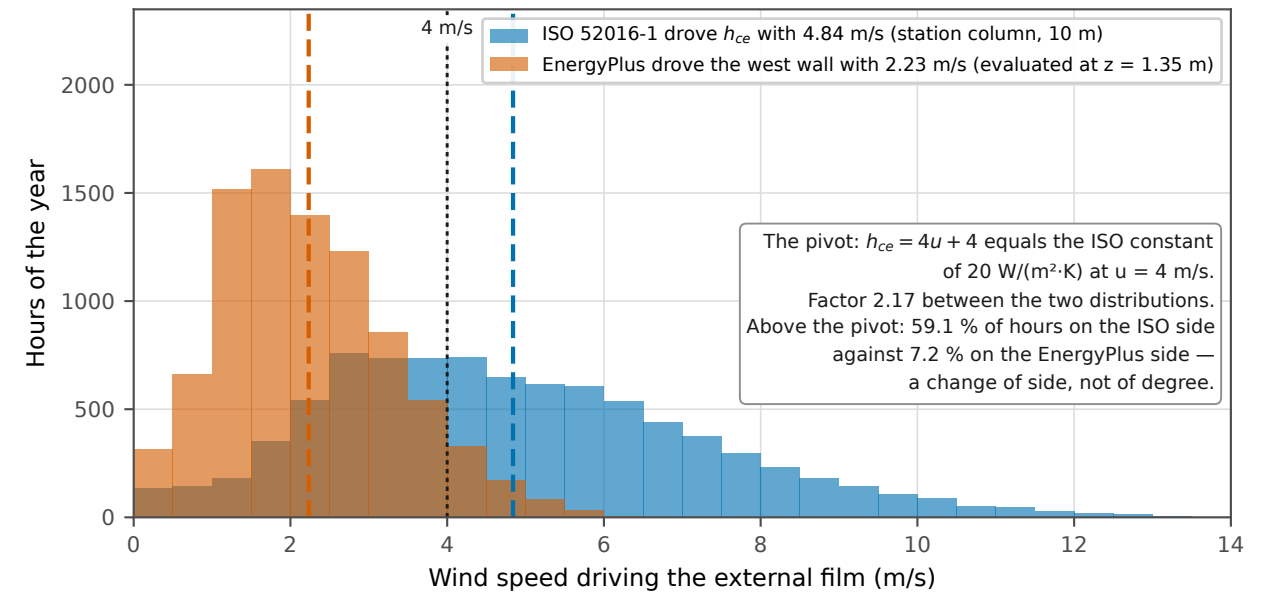
repaired state instrumented



Consequence: part of the same +961.73 kWh. Outdoor air attached to an ideal-loads system is delivered only while that system runs, and not at all while the zone free-floats in the deadband.

4 Wind at the wrong height

DEFECTIVE state instrumented



Consequence: on the canonical state, heating -0.86 kWh and cooling +6.48 kWh. The C2 correction's effect on cooling reverses sign, -4.73 → +1.75 kWh. Resolving this also revealed the terrain-correction mismatch of Section 4.4.

All four produced plausible annual totals and raised no warning. The reference ran to completion, its severe-error count was zero, and its annual heating of 1,120.2 kWh is an unremarkable number for this dwelling — it was wrong by 962 kWh, and the only way to see that was to look at the hourly streams. A validation that compares annual totals cannot detect any of these four, which is the point the paper makes about detailed reference models.